

Exploring Battle Dynamics Through Lanchester's Differential Equations

Limitations and Model Improvements

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Project Aim

- Study the classical Lanchester combat model.
- Identify its limitations: no space, no movement, no randomness, homogeneous forces.
- Build a 2D spatial model using a grid of cells.
- Convert the spatial problem into a system of ODEs.
- Simulate several tactical scenarios in R.

Classical model $\longrightarrow$ Spatial discretization $\longrightarrow$ Numerical simulation

Classical Lanchester Combat Model

Let $A(t)$ and $B(t)$ be the strengths of two opposing forces.

$$\frac{dA}{dt} = -\beta B, \quad \frac{dB}{dt} = -\alpha A,$$

where $\alpha, \beta > 0$ are combat effectiveness parameters.

A useful invariant is obtained by eliminating t :

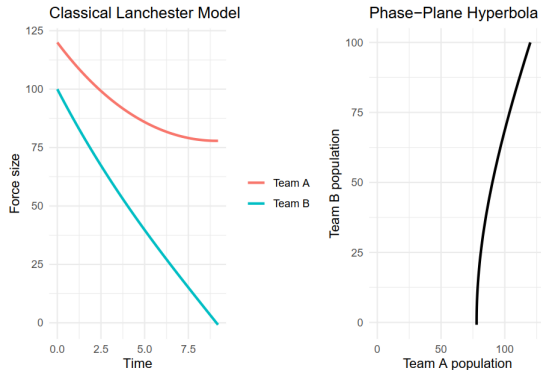
$$\frac{dA}{dB} = \frac{\beta B}{\alpha A} \implies \alpha A^2 - \beta B^2 = C.$$

Therefore the winning condition is:

$$\alpha A_0^2 > \beta B_0^2 \Rightarrow A \text{ wins}, \quad \beta B_0^2 > \alpha A_0^2 \Rightarrow B \text{ wins}.$$

Plots

The plots below shows the gradual attrition of both forces. The side with the larger effective initial strength tends to survive.



The graph of *Team B* vs. *Team A* shows the idea of Lanchester's square law of collapse, where when the forces are large initially, they decrease at a linear rate with respect to each other. However, when they get smaller, the quadratic rate kicks in and one of the team collapses.

Figure: Classical Lanchester trajectories

Why the Classical Model Needs Improvement

The standard model is elegant, but it assumes:

- forces are spatially mixed,
- effectiveness is constant,
- no reinforcement enters the battlefield,
- all troops are homogeneous,
- there is no movement or terrain effect,
- the outcome is purely deterministic.

These assumptions are too restrictive for realistic battle dynamics.

Need: a model with space + movement + local combat

2D Spatial Lanchester Model: Why Discretization is Used

A full PDE model is mathematically heavy.

So the battlefield is divided into an $n \times n$ grid:

$$A_{ij}(t), B_{ij}(t), \quad i, j = 1, \dots, n.$$

Each cell behaves like a small battle zone.

Discrete spatial model:

$$\frac{dA_{ij}}{dt} = -\beta B_{ij} + \text{Inflow}_{ij}^A - \text{Outflow}_{ij}^A + R_{ij}^A(t),$$

$$\frac{dB_{ij}}{dt} = -\alpha A_{ij} + \text{Inflow}_{ij}^B - \text{Outflow}_{ij}^B + R_{ij}^B(t).$$

This turns the spatial problem into a large coupled ODE system.

Algorithm of the R Code for the 2D Model

The R implementation follows this algorithm:

- ➊ **Index the grid:** convert cell coordinates (i, j) into a vector index.
- ➋ **Define movement:** choose a direction for each team (east, west, north, south, diagonal, or stay).
- ➌ **Create reinforcements:** add source terms at boundary cells or chosen locations.
- ➍ **Build the RHS function:** compute inflow, outflow, attrition, and sources.
- ➎ **Solve numerically:** use `ode()` from `deSolve`.
- ➏ **Stop at extinction:** truncate the solution when one team becomes effectively zero.
- ➐ **Visualize:** plot totals over time and heatmaps of cell densities.

Key functions used:

`cell_index()`, `make_next_map()`, `make_source()`, `make_spatial_rhs()`, `truncate_solution()`

Simulation Scenarios in R

Four scenarios are explored on a 4×4 grid:

- **Opposite boundaries:** both teams attack toward the middle.
- **Opposite corners:** diagonal confrontation.
- **Attacker vs defender:** one side advances while the other holds position.
- **Both defending:** both teams move away from the center.

Typical outputs:

- total force versus time,
- final heatmaps for each team,
- comparison of tactical outcomes.

These plots reveal how movement changes the battle outcome.

Plots

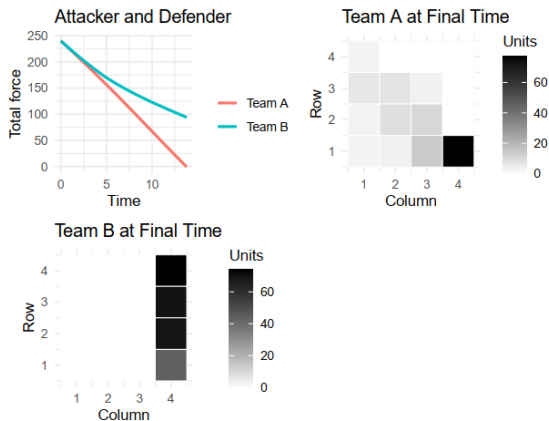


Figure: Example of Spatial Lanchester trajectories

(There are certain numerical limitation to this.)

Local Invariant and Winning Conditions

If movement is negligible in a single cell, then that cell behaves like the classical model:

$$\frac{dA_{ij}}{dt} = -\beta B_{ij}, \quad \frac{dB_{ij}}{dt} = -\alpha A_{ij}.$$

So locally,

$$\alpha A_{ij}^2 - \beta B_{ij}^2 = C_{ij}.$$

This is called a **local invariant** because it stays constant only in a local clash zone.

For the full spatial model:

- exact global invariants usually disappear,
- movement redistributes troops across cells,
- winning conditions become harder to write in closed form,
- but approximate front-line square-law behavior may still appear.

Limitations and Further Directions

The spatial ODE model is still simplified. Possible extensions:

- stochastic casualty rates,
- heterogeneous troop classes,
- terrain-dependent movement,
- morale and fatigue,
- delayed response,
- calibration with real or simulated data.

A more general framework could be:

$$\frac{dA_{ij}}{dt} = -\beta_{ij}(t)B_{ij} + m_{ij}^A(t) + r_{ij}^A + \epsilon_{ij}^A$$

$$\frac{dB_{ij}}{dt} = -\beta_{ij}(t)A_{ij} + m_{ij}^B(t) + r_{ij}^B + \epsilon_{ij}^B.$$

where ϵ_a, ϵ_b represent random effects.

- The classical Lanchester model gives a clean square-law winning condition.
- A 2D spatial extension captures movement and local clashes.
- Discretization converts the PDE idea into a solvable system of ODEs.
- The R algorithm simulates several battle scenarios on a grid.
- Local square-law behavior still helps interpret the model, even when a global invariant is lost.

Classical theory + Spatial discretization + R simulation = more realistic battle modeling

- Lanchester-Type Models of Warfare. Volume I&II (DTIC)
<https://apps.dtic.mil/sti/tr/pdf/ADA090843.pdf>
- Nigel Perry, *Fractal Effects in Lanchester Models of Combat*, 2009
<https://apps.dtic.mil/sti/html/tr/ADA523328/index.html>
- N. Cangiotti, M. Capolli, M. Sensi, *A Generalization of Lanchester's Model of Warfare*, 2022
<https://arxiv.org/abs/2112.06973>

Thank You

Questions and Discussion